INCH-POUND

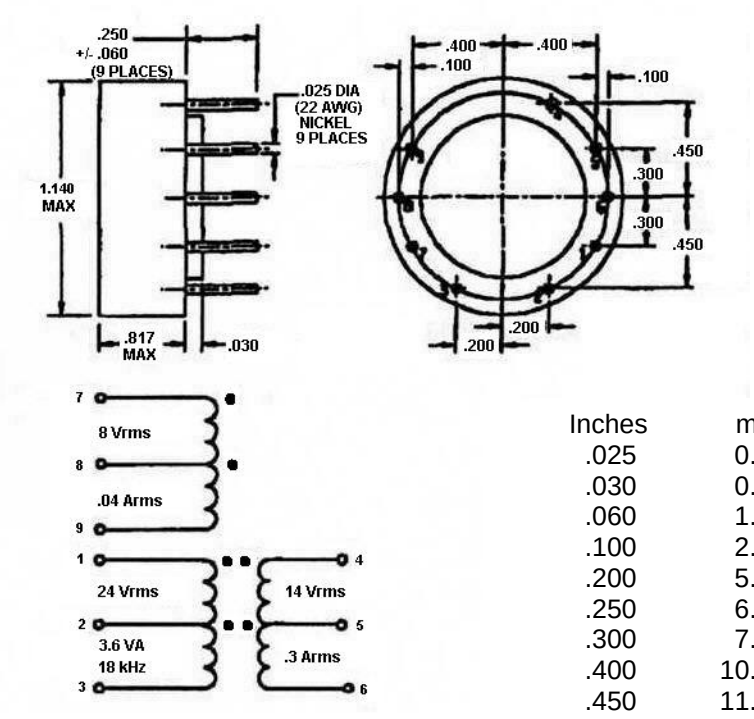
MIL-PRF-27/29C
28 April 2008
SUPERSEDING
MIL-PRF-27/29B
15 September 1975

PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, AUDIO FREQUENCY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.



Inches	mm
.025	0.64
.030	0.76
.060	1.52
.100	2.54
.200	5.08
.250	6.35
.300	7.62
.400	10.16
.450	11.43
.817	20.75
.880	22.35
1.140	28.96

WORKING VOLTAGE – 50 VOLTS

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configurations

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-3): 24 volts CT, 18 kHz.

Primary current (1-3): .15 amperes rms.

Secondary voltage: (4-6) 14 Volts rms CT.
(7-9) 8 Volts rms CT.

Secondary current: (4-6) .3 ampere rms.
(7-9) .04 ampere rms.

Working voltage: 50 volts.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle. Continuous.

Case: Encapsulated.

Material: Diallyl phthalate, ASTM-D5948, type SDG.

Terminals: Nickel, pin type.

Terminal height: .250 ±.060.

Weight: 4 ounces.

Altitude:

Operating: 10,000 feet.

Nonoperating: 40,000 feet.

Operating temperature range: -55°C to +105°C.

Maximum operating temperature: +105°C (ambient plus temperature rise).

Temperature rise: 20°C with 24 volts, 18 kHz across (1-3) at an ambient temperature of 85°C.

Terminal strength: Method 211, MIL-STD-202, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level): 500 volts rms (1-3) (4-6) (7-9).

Insulation resistance: Method 302 of MIL-STD-202, test voltage, 100 volts dc.

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Electrical characteristics:

No load: With 24 volts, 18 kHz across (1-3)

Current in (1-3): .15 ampere maximum.

Voltage across (4-6): 14.6 V maximum.

Voltage across (7-9): 8.4 V maximum.

Rated load: With 24 volts, 18 kHz across (1-3)

Current in (4-6): .3 ampere maximum.

Voltage across (4-6): 14 V ±2 percent.

Current in (7-9): .04 ampere maximum.

Voltage across (7-9): 8 V ±2 percent.

Polarity: Terminals 1, 4, and 7 shall be of the same instantaneous polarity.

Marking location: See figure 1.

Part or Identification Number (PIN): M27/29-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202
ASTM-D5948

Custodians:
Army - CR
Navy - EC
Air force – 11
DLA-CC

Preparing activity:
DLA - CC

(Project 5950-2008-006)

Review activities:
Army – AR, CR4
Navy - AS, MC
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.

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